

KISHAN DEKA

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EDUCATION

National Center for Nuclear Research, Warsaw **October 2022 – Present**
Doctoral Student (Supervisor : dr. hab. Paweł Bielewicz)

National Institute of Science Education and Research, Bhubaneswar **July 2017 – May 2022**
Integrated M.Sc. (5 years) in Physics (minor in Mathematics) **CGPA: 8.89/10**

RESEARCH INTERESTS

Weak lensing of Cosmic Microwave Background

Foreground contamination in lensing convergence reconstruction

Cross-correlation studies between lensing convergence and tracers of large-scale structures

PAST RESEARCH EXPERIENCES

PSF photometry methods for the WALOP instrument. **Aug 2021 – June 2022**
Guided by Dr. Tuhin Ghosh and Prof. A. N. Ramaprakash (Master's Thesis) *NISER Bhubaneswar*

- Use polar shapelet basis functions to model Point Spread Function (PSF) on different CCD positions.
- Computational modelling of PSFs for simulated star images and Robopol survey data.

Viscous Landau hydrodynamics in heavy-ion collisions **May 2019 – Oct 2019**
Guided by Dr. Amaresh Jaiswal (Report) *NISER Bhubaneswar*

- Studied heavy ion collisions considering non-conformal equation of state using Landau hydrodynamics.
- Worked on mathematical formulation of the problem and provided an analytic solution to the problem.

OTHER ACADEMIC PROJECTS

Tachocline confinement of solar differential rotation **May 2021 – Sept 2021**
Guided by Dr. Bidya Binaya Karak, IIT(BHU), Varanasi *Summer Project*

- Model the solar differential rotation profile by numerically solving the solar MHD equations.
- Add lorentz force term to check tachocline confinement of the differential rotation profile.

Signature of magnetic helicity in thermal dust polarization **July 2020 – May 2021**
Guided by Dr Tuhin Ghosh *Summer Project*

- Generate Stokes parameters maps for different existing galactic magnetic field models.
- Obtain power spectrum using XPol-Python package and check presence of non-zero TB correlation.

Classical Riemann problem and relativistic expansion of hot plasma. **Jan 2020 – June 2020**
Guided by Dr. Victor Roy, NISER Bhubaneswar *Semester Project (Report)*

- Studied semi-analytic solutions of 1D Riemann Problem and shock waves in classical MHD.
- Studied 1D relativistic expansion of hot magnetized plasma and reproduced some pre-existing results.

Existence and uniqueness results for BVPs in Sobolev spaces. **May 2018 – July 2018**
Guided by Dr. Bhupen Deka, IIT Guwahati *Summer Project (Report)*

- Studied various methods to solve ordinary and partial differential equations.
- Derived existence and uniqueness results for Dirichlet boundary value problem in Sobolev spaces.

PUBLICATIONS LIST

Journal article (ORCID : 0000-0002-8229-4326)

Deka, K. & Bielewicz, P., *Galactic foreground residual biases in CMB lensing convergence reconstruction and delensing of B-mode maps*, submitted to A&A, DOI: [10.48550/arXiv.2511.11147](https://doi.org/10.48550/arXiv.2511.11147)

Biswas, D., **Deka, K.**, Jaiswal, A., Roy, S., *Viscosity, nonconformal equation of state, and sound velocity in Landau hydrodynamics*, Phys. Rev. C **102**, 014912 (2020). DOI: [10.1103/PhysRevC.102.014912](https://doi.org/10.1103/PhysRevC.102.014912)

Maharana, S. et al., *Bright-Moon Sky as a Wide-Field Linear Polarimetric Flat Source for Calibration*, A&A **679**, A68 (2023). DOI: [10.1051/0004-6361/202346830](https://doi.org/10.1051/0004-6361/202346830)

Maharana, S. et al., *WALOP-South: A Four-Camera One-Shot Imaging Polarimeter for PASIPHAE Survey. Paper II – Polarimetric Modelling and Calibration*, J. of Astronomical Telescopes, Instruments, and Systems, **8**(3), 038004 (2022). DOI: [10.1051/0004-6361/202346830](https://doi.org/10.1051/0004-6361/202346830)

Conference proceeding

The CosmoVerse White Paper: Addressing observational tensions in cosmology with systematics and fundamental physics, Phys. of Dark Univ., **49**, 101965 (2025) . DOI: [10.1016/j.dark.2025.101965](https://doi.org/10.1016/j.dark.2025.101965)

Dynamics of QCD Matter – current status, International Journal of Modern Physics E, Vol. **30**, No. **02**, 2130001 (2021). DOI: [10.1142/s0218301321300010](https://doi.org/10.1142/s0218301321300010)

SEMINARS/TALKS

- *Optimal delensing of CMB B modes with stage-IV CMB and galaxy survey*, at XI Fundamental Cosmology in Santander, 2025 (selected talk)
- *Delensing with tomographic cross-correlation between LSST and CMB-S4 survey*, at LSST@Europe7, 2025 (selected talk)
- *Delensing of CMB B-modes with tomographic cross-correlation between LSST and CMB-S4 survey*, at DarkMap in Budapest, 2025 (selected talk)
- *Impact of galactic foregrounds in delensing of CMB B-mode polarisation*, CAMK PAN, Warsaw, 2025 (colloquium)
- *Impact of galactic foregrounds on delensing of CMB maps and detection of the primordial gravitational waves*, at Cosmoverse@Istanbul, 2025 (selected talk)
- *Galactic foreground bias in CMB lensing convergence reconstruction*, at Cosmology in Miramare, 2023 (selected talk)
- *PSF modeling for the WALOP instrument*, Astrophysics(BP4) seminar, NCBJ Warsaw (seminar)
- *Impact of the Galactic foregrounds on CMB lensing convergence reconstruction*, at EAS annual meeting, 2023, Krakow. (poster session)

TEACHING

- Teaching assistant in **Classical Electrodynamics** with prof. Andrzej Hryczuk, NCBJ, Warsaw, Fall 2025 (under-graduate level)

CONFERENCES/SCHOOLS

- XI Meeting on Fundamental Cosmology, Santander, Spain, Nov. 2025
- LSST@Europe, Poznan, Poland, Sept. 2025
- Advanced School on Precision Cosmology and Dark Matter, GGI, Florence, Italy, Aug. 2025 (school)
- DarkMap - Mapping the Dark Universe, Budapest, Hungary, Sept. 2025
- Cosmoverse@Istanbul, Turkey, June 2025

- PAiP-2025-Particle Physics in Poland, Warsaw, Poland, Feb. 2025
- Cosmoverse@Krakow, Poland, July 2024
- Cosmology in Miramare, at SISSA Trieste, Italy, Aug .2023
- EAS Annual meeting 2023, ICE Krakow, Poland, July 2023
- 9th IIST Astronomy and Astrophysics School, at IIST Thiruvananthapuram, Dec. 2020 (school)
- Dynamics of QCD Matter, at NISER Bhubaneswar, Aug. 2019
- Vijyoshi Science Camp, at IISER Kolkata, Dec, 2017 (winter school)

FELLOWSHIPS/AWARDS

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| • NAWA STER Fellowship by NAWA (NCN), Poland | 2023 – |
| • INSPIRE Fellowship by Dept. of Science and Technology, India | 2017 – 2022 |
| • Anundoram Baruah Award by Govt. of Assam, India | Dec 2015 |

RELEVANT SKILLS

Languages: Python(proficient), Fortran, C++
Packages: HEALPix, CAMB, COBAYA, GLASS, Plancklens.
Softwares/Tools: Mathematica, GNUplot, Latex.

SOCIAL INVOLVEMENT

Journal Club, Astrophysics division (BP4), NCBJ, Warsaw	2022-
NISER Astronomy Club, NISER Bhubaneswar	2017-2022
Science Activity Club, NISER Bhubaneswar	2018-2022

REFERENCES

Dr. hab. Paweł Bielewicz, NCBJ, Warsaw, Poland
 Dr. Tuhin Ghosh, NISER, Bhubaneswar, India
 Dr. Amaresh Jaiswal, NISER, Bhubaneswar, India